

NEUMAN ALKHALIL

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EXPERIENCE

Founding AI Software Engineer

Jul. 2025 - Present

[Horalix](#) | Sarajevo, Bosnia & Herzegovina

- Lead AI development of a platform that automatically analyzes heart ultrasound exams and assists clinicians with measurements and reporting, currently being tested and validated across 3 hospital sites.
- Designed and integrated a multi-model computer vision and deep learning pipeline for cardiac measurements, image-view recognition, Doppler analysis, and automated clinical reporting, with reliability and fallback logic for hospital use.
- Built and deployed the production AI system using Python, FastAPI, PyTorch, React, PostgreSQL, Docker, DICOM/Orthanc, and GPU inference, supporting both on-premise clinical deployments and AWS cloud infrastructure.

Software Engineer

Mar. 2026 – Jun. 2026

Authority Partners | Sarajevo, Bosnia & Herzegovina

- Developed a full-stack internal platform using C#, ASP.NET Core, Entity Framework Core, PostgreSQL, Angular, and Keycloak, implementing authentication, role-based workflows, and REST APIs.
- Built an AI recruitment chatbot backed by a cached knowledge base of open positions and job descriptions, enabling candidates to discuss roles, submit resumes, and interact with recruiting workflows through natural language.
- Integrated the chatbot with Authority Partners' existing CATS recruiting system and a newly developed ATS through Model Context Protocol (MCP), with HubSpot contact creation and deployment across staging and production using Azure DevOps, CI/CD, Bicep, and SonarCloud.

Software Developer & IT Specialist

Oct. 2021 - Dec. 2021

SARAJEVO UNLIMITED | Sarajevo, Bosnia & Herzegovina

- Developed and maintained a full-stack event platform supporting 500+ participants and 30+ keynote speakers, while providing technical support for 20+ live presentations.

Software Engineer Intern

Sept. 2021 - Nov. 2021

ZIRA | Sarajevo, Bosnia & Herzegovina

- Built a full-stack application using Angular, Spring Boot, PostgreSQL, and REST APIs, with live RSS data, real-time querying, and database optimizations.

RELEVANT PROJECTS

[LPX.GG](#) — League of Legends Analytics Platform

Astro, React, TypeScript, Cloudflare Workers, Hono, PostgreSQL, Drizzle ORM, Riot Games API

- Built and launched a live full-stack analytics platform for player profiles, match history, builds, and performance data, integrating the Riot Games API with caching and PUUID-based player identity.
- Designed a scalable, SEO-focused architecture using server-rendered pages, Cloudflare Workers, and PostgreSQL-backed data persistence.

[Real-Time Speech Emotion Recognition](#) — Speech AI

Python, Whisper, WavLM, WebRTC, Silero VAD, RxPY

- Developed a real-time speech analysis pipeline combining transcription, vocal emotion recognition, and text sentiment analysis, with asynchronous audio processing and cross-dataset model evaluation on RAVDESS and IEMOCAP.

[GAN-Based Image Restoration](#) — Computer Vision

PyTorch, GANs, U-Net, CNN

- Developed a GAN-based image restoration system with modified generator and discriminator architectures to reconstruct degraded images across multiple noise types while also supporting image colorization.

TECHNICAL SKILLS

Languages: Python, C#, TypeScript, JavaScript, SQL, Java

AI & Machine Learning: PyTorch, Scikit-learn, Computer Vision, LLMs, NLP, CUDA, Model Deployment, Model Context Protocol (MCP)

Backend & Data: FastAPI, ASP.NET Core, Entity Framework Core, PostgreSQL, SQLAlchemy, REST APIs

Frontend: React, Angular, Astro, Electron

Cloud & DevOps: AWS, Docker, Terraform, Azure DevOps, CI/CD, Bicep, Cloudflare Workers, Keycloak, Git, SonarCloud

EDUCATION

Georgetown University

M.S. Computer Science | May 2024

Focus: AI & Machine Learning

University of Buckingham

B.S. Computer Science | Jul 2022

CERTIFICATES

- Accelerated Computing with CUDA
NVIDIA | 2024

AWARDS

- MedReCon Hackathon Winner
SSST | 2026
- Discover ViennaUp Winner
Vienna Business Agency | 2026
- FutureMinds Lab Healthcare Winner
SSST/GIZ | 2025
- Adria Future Hackathon Winner
Adria Future Summit | 2026

LANGUAGES

- Fluent - English
- Intermediate – Bosnian/Croatian

INTERESTS

- Travel
- Fitness
- Puzzles
- Competitive Strategy Games